

# Mesh Aspect Ratio and Cantilever-Beam Response

Two-person academic project | University of Windsor | 2025

Five Abaqus plane-stress models were compared with analytical beam-theory results to examine displacement convergence and stress sensitivity.

|           |                    |
|-----------|--------------------|
| 5         | Abaqus meshes      |
| 24 to 1.1 | aspect-ratio range |
| 45.92%    | coarse-mesh error  |
| 2.95%     | near-square error  |

## MODEL

Cantilever length 12 in; width 0.5 in; thickness 1 in;  $E = 30 \times 10^6$  psi; Poisson's ratio = 0.3; plane-stress formulation; total parabolic end load = 40,000 lb.

## METHOD

- Built meshes with element aspect ratios of 24, 6, 3.6, 1.5, and 1.1.
- Compared Abaqus tip displacement with the analytical target of -1.152 in.
- Reviewed maximum stress behaviour and interpreted shear-locking effects.

## EVIDENCE

Tip displacements progressed from -0.4993 in at AR 24 to -1.141 in at AR 1.1. The near-square mesh reduced displacement error to 2.95%; AR 3.6 produced the closest reported stress result.

## ENGINEERING FINDING

Element shape materially affected the response. Near-square elements improved displacement convergence, while stress accuracy did not follow the same simple monotonic pattern.

## LIMITATIONS

The comparison is specific to the selected plane-stress element formulation, geometry, boundary conditions, and loading definition.

Evidence basis: Abaqus models, course report, analytical checks, and MATLAB calculations. This summary reports academic verification work and does not represent a physical test.